Project 23-Link: Boundary Conditions of the $(3+1)^n$ Model

Aymane Amri Amine
Pre-Med Candidate | Biomedical Engineering Simulation
Focus: Advanced Physiopathology & Genomic Systems

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1 Introduction

The classical Mendelian polyhybrid ratio, defined by $(3:1)^n$, assumes perfect independent assortment. This project explores the transition from discrete Mendelian ratios to quantitative genomics by simulating the impact of the 23 human chromosome pairs and physical linkage.

2 Mathematical Framework

For n independent loci, the probability P of an individual possessing k dominant phenotypes follows a **Binomial Distribution**:

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k} \tag{1}$$

Where:

- $n = 23$ (Number of loci, representing the chromosomal scale).
- $p = 0.75$ (Probability of dominance in an $Aa \times Aa$ cross).

3 The Gaussian Transition

As n increases, the Central Limit Theorem dictates that the distribution converges toward a **Normal Distribution** $\mathcal{N}(\mu, \sigma^2)$:

$$\mu = np = 23 \times 0.75 = 17.25 \tag{2}$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{23 \times 0.75 \times 0.25} \approx 2.07 \tag{3}$$

This convergence demonstrates that in physiopathology, polygenic traits appear as a continuous spectrum rather than discrete categories. The probability of observing a total recessive phenotype ($k = 0$) becomes infinitesimal:

$$P(X=0) = (0.25)^{23} \approx 1.42 \times 10^{-14} \tag{4}$$

4 Physical Constraints: Linkage Disequilibrium

The model $(3+1)^n$ fails when loci are physically linked on the same chromosome. In our simulation, the introduction of **Haplotype Blocks** (where $Cov(X_i, X_j) \neq 0$) reduces the effective entropy of the system.

Instead of a smooth Gaussian curve, the population polarizes into specific clusters, a phenomenon known as **Linkage Disequilibrium (LD)**. This explains the clustering of drug responses in pharmacogenomics.

5 Conclusion

Project 23-Link proves that the 23-chromosome structure of the human genome acts as a physical ceiling for Mendelian randomness. The transition from $n = 1$ to $n = 23$ marks the shift from classical genetics to the quantitative reality of modern clinical practice.